

SUPRIYO CHAUDHURI

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COURSEWORK / SKILLS

- | | | | |
|--------------------------------|-------------------------------------|----------------------------|--------------------------------------|
| • Data Structures & Algorithms | • Database Management System (DBMS) | Programming | • Global Navigation Satellite System |
| • Operating Systems | • Artificial Intelligence | • OOPS Concept | • Technical Communication |
| • Network Security | • Advanced Python | • Advanced Web Development | |

INTERNSHIP

Space Applications Centre [ISRO] [🔗](#)

May 2026 – July 2026

SWE Intern

Ahmedabad, Gujarat

- Contributed to the development of **PulsarNav.ai**, a **GPS-independent autonomous deep-space navigation platform** leveraging pulsar timing measurements and direction Vectors for spacecraft localization and its state estimation.
- Designed and implemented **navigation algorithms**, **signal-processing pipelines**, and mathematical estimation models using the **NANOGrav 12.5-year pulsar timing dataset** to simulate and validate pulsar-based spacecraft navigation performance.
- Executed autonomous spacecraft navigation algorithms inspired by the mathematical framework presented in **Navigation in Space by X-ray Pulsars** by **Amir Abbas Emadzadeh** and **Jason L. Speyer**, including **pulse delay estimation**, **epoch folding**, **recursive state estimation**, and **pulsar navigation geometry** for spacecraft localization.
- Developed mathematical models for **pulsar timing**, **navigation geometry**, **least-squares estimation**, and **Monte Carlo simulations** to evaluate spacecraft localization accuracy, quantify estimation uncertainty, and assess navigation robustness under varying timing noise and mission scenarios.
- Enhanced scalable **simulation**, **testing**, and **validation frameworks** to benchmark navigation accuracy, evaluate algorithm robustness, and analyze spacecraft localization performance across diverse deep-space mission scenarios.
- Streamlined **navigation algorithms** through modular software design, efficient implementation, debugging, and computational performance analysis to improve reliability, maintainability, and execution efficiency.
- Followed collaborative **software engineering practices** using Git-based version control, modular development, iterative testing, and documentation to build reliable research-grade software for autonomous spacecraft navigation.

PROJECTS

Expense Tracker [🔗](#) | [Node.js](#), [Express.js](#), [PostgreSQL](#), [EJS](#), [JavaScript](#)

December 2025

- Developed a full-stack **Expense Tracker** using **Node.js**, **Express.js**, **PostgreSQL**, and **EJS**, following a modular and scalable software architecture.
- Applied secure **user authentication**, session management, and authorization to ensure reliable access control and data security.
- Designed an optimized **PostgreSQL** database with efficient **CRUD operations** and deployed responsive dashboards for expense tracking and financial analytics.
- Built and Enhanced **RESTful APIs**, improving backend performance through debugging, database query optimization, and clean software design.
- Applied **Agile development methodologies** and modern **software engineering practices**, leveraging **Git**-based version control, modular development, iterative testing, and collaborative code management to deliver a reliable and maintainable application.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL, NoSQL, R, XML, Go

Developer Tools: VS Code, jupyter notebook, Antigravity, Vercel, Ollama, Obsidian, IntelliJ Idea Ultimate

Technologies/Frameworks: GitHub, ReactJS, Redis, NextJS, NodeJS, ExpressJS, Git, MongoDB, PostgreSql, Bootstrap, Flutter

EDUCATION

Pandit Deendayal Energy University

July 2023 – July 2026

B.tech in Computer Engineeering - 3.42 - 4.00

Gandhinagar, Gujarat